

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Interconnection of Large Loads to the)
Interstate Transmission System)

Docket No. RM26-4-000

**REPLY COMMENTS OF
AMERICAN ELECTRIC POWER SERVICE CORPORATION**

American Electric Power Service Corporation (“AEP”), on behalf of its affiliates,¹ respectfully submits the following reply comments in response to comments filed in the above-captioned proceeding regarding the Advanced Notice of Proposed Rulemaking (“ANOPR”) pursuant to the Notice of Request for Comments (“Notice”) issued by the Federal Energy Regulatory Commission (the “Commission”) on October 27, 2025.²

At present, the generation interconnection process is too slow to provide generation capacity for large loads and win the artificial intelligence race,³ match pace with economic development, and ensure national security. More generation resources and more transmission infrastructure are desperately needed to ensure the safe and reliable interconnection of and service to new large loads. As noted in AEP's Initial Comments, a root cause of the problem is the inability of the current, siloed processes to drive the construction of needed generation and transmission

Appalachian Power Company, Indiana Michigan Power Company, Kentucky Power Company, Kingsport Power Company, Ohio Power Company, Wheeling Power Company, Public Service Company of Oklahoma, Southwestern Electric Power Company, AEP Appalachian Transmission Company, Inc., AEP Indiana Michigan Transmission Company, Inc., AEP Kentucky Transmission Company, Inc., AEP Ohio Transmission Company, Inc., AEP West Virginia Transmission Company, Inc., AEP Oklahoma Transmission Company, Inc., and AEP Southwestern Transmission Company, Inc.

² On November 7, 2025, the Commission extended the deadline for comments to November 21, 2025 and for reply comments to December 5, 2025. Notice Granting Extension of Time, Docket No. RM26-4 (issued Nov. 7, 2025).

³ See, The White House, *Winning the Race America's AI Action Plan* (July 2025), <https://www.whitehouse.gov/wp-content/uploads/2025/07/Americas-AI-Action-Plan.pdf>.

fast enough.⁴ Without much needed reforms, the current approaches threaten to produce insufficient and inefficient infrastructure to reliably support large load growth and economic development. Left unaddressed, this could result in potentially unsustainable price increases and unaffordable energy bills that do not protect existing customers. To protect other customers, large loads must pay their fair share for the transmission system expansion needed to serve them, and the Commission also needs to fundamentally reconsider how generator interconnection and transmission system planning are conducted to ensure that reforms drive cost-effective development to support interconnecting loads.

I. EXECUTIVE SUMMARY

There is broad agreement among the ANOPR commenters that reforms are needed to improve the timeliness, predictability, and transparency of large load interconnection processes.⁵ There also is broad agreement that any Commission action should not disrupt the current, productive efforts by utilities working with their respective states, RTO/ISOs, and large load customers to integrate large loads.⁶ Rather, any processes adopted by the Commission should

⁴ Comments of Am. Elec. Power Serv. Corp., Docket No. RM26-4, at 4, 15 (filed Nov. 21, 2025) (“AEP Initial Comments”).

⁵ Comments of the Am. Clean Power Ass’n, Docket No. RM26-4, at 1 (filed Nov. 21, 2025) (“American Clean Power Comments”); Comments of Oracle Am., Inc., Docket No. RM26-4, at 1 (filed Nov. 21, 2025) (“Oracle Comments”); Comments of PSEG Cos., Docket No. RM26-4, at 2 (filed Nov. 21, 2025) (“PSEG Comments”).

⁶ *See, e.g.*, Initial Comments of Amazon Energy LLC, Docket No. RM26-4, at 3 (filed Nov. 21, 2025) (“Amazon Energy LLC Comments”); Initial Comments of the Env’t Law & Pol’y Ctr., Docket No. RM26-4, at 2–5 (filed Nov. 21, 2025) (“ELPC Initial Comments”); Comments of Indicated PJM TOs, Docket No. RM26-4, at 16–17 (filed Nov. 21, 2025) (“PJM TOs Comments”); Comments of the Large Public Power Council, Docket No. RM26-4, at 3–5 (filed Nov. 21, 2025) (“LPPC Comments”); Comments of the MD Energy Admin., Docket No. RM26-4, at 3 (filed Nov. 21, 2025) (Maryland Energy Administration Comments); Initial Comments of Louisiana & Mississippi Pub. Serv. Comm’n, Docket No. RM26-4, at 4 (filed Nov. 21, 2025) (LPSC/MPSC Comments); Initial Comments of MISO Transmission Owners, Docket No. RM26-4, at 33–34 (filed Nov. 21, 2025) (“MISO TOs Comments”); Comments of Consumer Protection Off. of the Ohio Consumers’ Counsel, Docket No. RM26-4, at 1, 4 (filed Nov. 21, 2025) (“Ohio OCC Comments”); Comments of Southwest Power Pool Transmission Owner Grp., Docket No. RM26-4,

complement those effective processes, while also advancing protections for existing consumers.⁷ AEP's proposal in its Initial Comments achieves this goal by focusing on long-term, proactive solutions to generator interconnection, transmission planning, and cost allocation. AEP's proposal supports the expedited connection of data centers and large industrial loads to the electric grid by effectively combining short-term speed-to-power solutions, with longer term generator interconnection reforms and proactive transmission planning, which address the root causes of the current slow pace of large load interconnections. AEP's proposal provides the most effective way to address issues created by the interconnection of large load holistically, quickly and without disrupting existing processes that have been put in place in the various states and RTO/ISOs.

Any reforms adopted by the Commission as part of this ANOPR process should focus on long-term, proactive solutions to develop multi-value, high voltage transmission, supportive national cost allocation, and the interconnection of much needed new generation. This is where the Commission can have the greatest, most-effective, and least-disruptive impact to advance large load interconnections. The Commission can supplement these efforts with short-term solutions to improve speed-to-power in the near term, but the Commission must be selective about any such short-term solutions. The Commission should be mindful that any short-term solution does not threaten reliability or derail current efforts at the state and regional level to effectively and efficiently interconnect large loads. Additionally, the Commission should thoughtfully consider

at 4-5 (filed Nov. 21, 2025) ("SPP TOs Comments"); Comments of Vantage Data Center, Docket No. RM26-4, at 2-3 (filed Nov. 21, 2025) ("Vantage Data Center, Comments").

⁷ Leaving current state and RTO/ISO processes in place also avoids disputes regarding state jurisdiction over retail service to large loads and other end use customers, which could impede the implementation of reforms that would otherwise accelerate the provision of service to large loads. Throughout these comments, AEP intends to advocate for implementation of reforms and solutions within FERC's jurisdiction in a manner that does not displace the retail jurisdiction of state commissions. It is assumed by these comments that all loads will be served by load serving entities ("LSE") who will take wholesale transmission service from transmission providers and owners to support their retail service to loads.

how any actions in this proceeding will protect existing customers from bearing disproportionate costs of large load infrastructure development. As discussed in AEP's Initial Comments, a national transmission cost allocation can lower costs by expediting investment in a proactively-planned national transmission backbone, providing nationwide access to a cost-effective and efficient grid.⁸

The record provides a few promising proposals with elements that the Commission should consider promoting or adopting on a national level, such as certain elements of the Southwest Power Pool's ("SPP's") High Impact Large Load ("HILL") and High Impact Large Load Generation Assessment ("HILLGA") proposals⁹ and NRG's open season proposal,¹⁰ if implemented in a manner that complements more comprehensive, proactive transmission planning. Critically, the Commission should focus on the necessary long-term solutions to allow system operators to efficiently and cost effectively manage the necessary build out to support the large load additions that are necessary to win the AI race.

II. THE COMMISSION SHOULD NOT DISRUPT PROGRESS AT THE STATE AND REGIONAL LEVEL

A. Any Commission action should be principles-based, avoid prescriptive requirements, and allow for regional flexibility

Any rules adopted by the Commission in this proceeding should be principles-based, avoid prescriptive requirements, and allow for regional flexibility. As many commenters note, states are already working with their state regulated utilities and large load customers in their respective

⁸ See AEP Initial Comments at 23; *see infra* section IV.A, nn.48-49 (noting comments that describe how the addition of large loads can lower costs for existing customers).

⁹ See Comments of Southwest Power Pool, Inc. Docket No. RM26-4, at 4-6 (filed Nov. 21, 2025) ("SPP Comments").

¹⁰ Initial Comments of NRG Energy, Docket No. RM26-4, at 17 – 26 (filed Nov. 19, 2025) ("NRG Comments").

states to advance interconnection processes that work for their specific circumstances.¹¹ The majority of commenters strongly agree that any Commission reforms to address large load interconnection should avoid disrupting these efforts that are already in progress.¹² Rather, the Commission's actions should allow for regional flexibility, avoiding prescriptive national requirements that could, in practice, hinder large load interconnections. The U.S. Chamber of Commerce noted that any Commission action here "should be mindful to not upend the progress

¹¹ See Comments of Wash. Utils. & Transp. Comm'n, Docket No. RM26-4, at 1 (filed Nov. 24, 2025) ("Washington UTC Comments") (ANOPR policies may conflict with the ongoing work of Washington's Data Center Working Group established by Governor Ferguson's Executive Order 25-05); Comments of Ind. Energy Ass'n, Docket No. RM26-4, at 2 (filed Nov. 24, 2025) ("Indiana EA Comments") (Indiana has demonstrated that it is proactively and effectively addressing large load customer issues, through legislation, unique tariffs, and expedited regulatory proceedings); Initial Comments of the Cal. Pub. Utils. Comm'n, Docket No. RM26-4, at 2 (filed Nov. 21, 2025) ("California PUC Comments") (CPUC and other states are already working on state-jurisdictional tariffs and standardized rules for large load energization); Comments of Pub. Staff – N.C. Utils. Comm'n, Docket No. RM26-4, at 5-6 (filed Nov. 21, 2025) ("North Carolina UC Staff Comments") (North Carolina UC has proactively addressed the challenges, including conducting a technical conference on fairly and efficiently serving large load customers); Initial Comments of the Wis. Elec. Power Co. et al. Docket No. RM26-4, at 4-6 (filed Nov. 21, 2025) ("WEC Comments") (discussing Very Large Customer and Bespoke Resources Tariffs, which together address large loads in Wisconsin Electric's service territory); Comments of Pub. Util. Comm'n of Ohio's Fed. Energy Advocate, Docket No. RM26-4, at 2-3 (filed Nov. 21, 2025) ("PUCO FEA Comments") (PUCO has already begun working with regulated utilities in Ohio to create an orderly and expedited process by which new data center loads can request interconnection studies and an electric service agreement).

¹² See e.g., Amazon Energy LLC Comments, at 3 (noting that "[l]arge load developers have expended significant time and resources developing new large load projects . . . and made substantial investments in reliance on existing interconnection rules and procedures[.]" reforms that threaten that reliance could "cause substantial delay, or even force withdrawals, thereby undermining the goals . . . that the ANOPR seeks to advance."); Comments of Data Center Coalition, Docket No. RM26-4, at 4 (filed Nov. 21, 2025) ("DCC Comments") ("Reforms intended to improve timeliness and predictability should not inadvertently disrupt projects that have already made significant financial, contractual, and engineering commitments"); Comments of Eolian, L.P., Docket No. RM26-4, at 23 (filed Nov. 21, 2025) ("Eolian Comments") ("Any load interconnection customer that has commenced a state jurisdictional process consistent with such a rule should then have the option to continue to be processed pursuant to that process even after acceptance of the applicable compliance filing, in order to limit disruption to the orderly processing of large load interconnections."); Comments of Am. Pub. Power Ass'n, Docket No. RM26-4, at 4 (filed Nov. 21, 2025) ("APPA Comments") (the Commission should not undermine the "efforts of retail regulators" to provide through retail tariffs the "clarity to data center developers" which have "help[ed] create positive environment for investment"); Ohio OCC, at 4 ("FERC's rule must not undercut or preempt state efforts to manage large load growth through retail tariffs and rate design."); LPPC Comments, at 6 (the Commission must ensure "it does not disrupt efforts now being undertaken by state authorities to deal with the urgency of serving this new load while protecting existing customers.").

that is currently being made to interconnect large load across the country.”¹³ Many state regulators noted that they are already working to address large load interconnection issues at the state level, and the Commission should avoid disrupting this progress.¹⁴ Large load customers also expressed their strong desire for collaboration and coordination between federal and state regulators, and that the Commission should not interrupt in-progress projects with one-size-fits-all nationwide requirements.¹⁵ Thus, any reforms that the Commission implements should be principles-based and should avoid prescriptive requirements that will interfere with state and regional advancements on large load interconnection.

AEP’s recent retail data center tariffs are a primary example of the success of current state-level reforms. Working with state regulators in Ohio¹⁶ and across its multi-state footprint,¹⁷ AEP has crafted state-level large load tariffs to accelerate interconnection of these customers to the system while protecting the interests of all customers, a critical component of any action the Commission may choose to take. Several commenters highlight the success of AEP’s state-level

¹³ Comments of U.S. Chamber of Com., Docket No. RM26-4, at 2 (filed Nov. 21, 2025) (“U.S. Chamber of Commerce Comments”).

¹⁴ See, e.g., Comments of Pennsylvania Pub. Utils. Comm’n, Docket No. RM26-4, at 8 (filed Nov. 21, 2025) (PA PUC Comments); Initial Comments of Louisiana & Mississippi Pub. Serv. Comm’n, Docket No. RM26-4, at 6-8 (filed Nov. 21, 2025) (LPSC/MPSC Comments); PUCO FEA Comments, at 2-3; Initial Comments of Va. State Corp. Comm’n, Docket No. RM26-4, at 4-5 (filed Nov. 21, 2025) (“Virginia SCC”); Comments of N.Y. State Pub. Serv. Comm’n, Docket No. RM26-4, at 7 (filed Nov. 21, 2025) (“NY PSC Comments”).

¹⁵ Comments of META Platforms, Inc., Docket No. RM26-4, at 3 (filed Nov. 21, 2025) (“META Comments”) (“The Commission ... should not rush to impose a one-size-fits-all regulatory solution nationwide.”); Initial Comments of Microsoft, Inc., Docket No. RM26-4, at 8 (filed Nov. 21, 2025) (“Microsoft Comments”) (supporting collaboration between federal and state regulators); Amazon Energy LLC Comments at 3-4 (arguing the Commission should not disrupt large load projects that are already in progress).

¹⁶ *In re Application of Ohio Power Co. for New Tariffs Related to Data Centers & Mobile Data Ctrs.*, PUCO Case No. 24-508-EL-ATA, Opinion and Order (July 9, 2025).

¹⁷ In addition to Ohio, AEP has implemented these tariffs in Indiana, West Virginia, and Kentucky and has filed similar tariff structures for state commission approval in Virginia, Texas, and Michigan. AEP Initial Comments at 6, n.8.

tariffs, identifying them as an example of effective state-level solutions for large load interconnection.¹⁸ The Public Utility Commission of Ohio (“PUCO”) Federal Energy Advocate highlighted the AEP Ohio Tariff as an example of the PUCO’s work with Ohio utilities to “create an orderly and expedited process” that data center loads can use to connect to the system. Since implementation of the Tariff on July 11, 2025, “36 data center cites representing over 13,000 MW of new load have filed applications” to interconnect.¹⁹ The AEP Ohio tariff also received support from the Ohio Consumers Counsel, stating it “represent[s] careful policy judgements about economic development, cost recovery, and local grid capacity” and “ensures that local customers are not at a disadvantage in any large industrial projects.”²⁰ The Commission should build on this important progress at the state level that balances the need for speed-to-power with the need to protect existing customers.²¹

At the same time, RTO/ISOs are actively working to improve processes for generator interconnection and transmission development to enhance speed-to-power. The Commission should not overlook these reforms, which are the result of stakeholder-driven, consensus processes that reflect the unique needs of specific regions. For example, PJM has an ongoing stakeholder process to develop reforms. PJM is working to implement a process that addresses large load

¹⁸ PUCO FEA Comments at 2; Ohio OCC Comments at 4, 8; Comments of the AES Corp., Docket No. RM26-4, at 5 n.7 (filed Nov. 21, 2025) (“AES Comments”); Mot. to Intervene and Comments of DTE Elec. Co., Docket No. RM26-4, at 16 (filed Nov. 21, 2025) (“DTE Comments”).

¹⁹ PUCO FEA Comments at 2.

²⁰ Ohio OCC Comments at 4, 8.

²¹ AEP disagrees with Vistra Corp.’s comments that advocate for the Commission to interfere with the AEP Ohio Tariff. Vistra provides an annotated version of the AEP Ohio Tariff to illustrate its argument that if the Commission asserts jurisdiction over the interconnection of large loads, it will leave state efforts undisturbed. But the Vistra’s annotations to the AEP Ohio Tariff actually demonstrate significant disruption to AEP’s and Ohio’s efforts, stripping out material provisions. The Commission should not disrupt utility and state regulator efforts in this way. *See* Comments of Vistra Corp., Docket No. RM26-4, at 15, Exhibit A (filed Nov. 21, 2025) (“Vistra Comments”).

interconnection while maintaining system reliability, with the aim of gaining broad stakeholder support.²² And recently, as described in more detail below, SPP filed new rules and tariff provisions, with utility and customer support, to govern aspects of the large load interconnection process.²³ Here again, the Commission should ensure that any processes developed through this ANOPR proceeding are additive and do not disrupt any regional progress.

B. Large load interconnection network upgrade costs should be rolled into transmission rates

AEP agrees with commenters arguing that network upgrade costs for large load interconnections should be rolled into transmission rates and that direct assignment of 100% of such network upgrade costs to large load customers is not appropriate.²⁴ As many commenters discussed, the Commission’s longstanding precedent is that network upgrades benefit the entire system and therefore their costs should be rolled into transmission providers’ rates and charged to all system customers.²⁵ As these commenters noted, this makes sense—large loads should pay their fair share for use of the transmission system and not just for the costs necessary to

²² PJM Interconnection, L.C.C., *PJM Board of Managers Letter to Stakeholders* (Aug. 8, 2025), <https://www.pjm.com/-/media/DotCom/about-pjm/who-we-are/public-disclosures/2025/20250808-pjm-board-letter-re-implementation-of-critical-issue-fast-path-process-for-large-load-additions.pdf> ; PJM Interconnection, L.C.C., *Large Load Additions Stakeholder Feedback* (Sept. 15, 2025), <https://www.pjm.com/-/media/DotCom/committees-groups/cifp-lla/2025/20250915/20250915-item-04---stakeholder-feedback-on-large-load-additions---pjm-presentation.pdf>.

²³ Sw. Power Pool, Inc., *Southwest Power Pool board approves accelerated pathway for large load connection* (Sept. 16, 2025), <https://www.spp.org/news-list/southwest-power-pool-board-approves-accelerated-pathway-for-large-load-connection/>.

²⁴ See, e.g., Comments of WIRES, Docket No. RM26-4, at 6 (filed Nov. 21, 2025) (“WIRES Comments”); Comments of Edison Elec. Inst., Docket No. RM26-4, at 8-9 (filed Nov. 21, 2025) (“EEI Comments”); Comments of FirstEnergy Serv. Co., Docket No. RM26-4 at 6-7 (filed Nov. 21, 2025) (“FirstEnergy Comments”); Comments of Duke Energy, Docket No. RM26-4 at 22-23 (filed Nov. 21, 2025) (“Duke Comments”); PSEG Comments at 23-24; Joint Comments of Indus. Customer Orgs., Docket No. RM26-4 at 8-9 (filed Nov. 21, 2025) (“Industrial Customers Comments”).

²⁵ *Id.*

interconnect them to the system.²⁶ Also, as EEI noted, to protect existing customers from risks created by large loads, state-level reforms are in place that mitigate these risks and protect customers by setting minimum security requirements, among other tools.²⁷

Furthermore, it is fundamental that transmission owners earn a reasonable rate of return for owning and operating all transmission facilities, including those necessary to interconnect large load customers, consistent with long standing Commission policy for transmission upgrades integrated with the grid.²⁸ Transmission owners earn a lawful and approved return via their transmission rates by including the capital costs necessary to develop the network upgrades in their rate base. Thus, by including the network upgrade costs for large load interconnections in rolled-in transmission rates, transmission owners will be able to earn their approved return for owning and operating these facilities. This is critical to ensure that transmission owners can continue to attract the capital necessary to develop the transmission infrastructure needed to support the interconnection of large loads.

²⁶ See EEI Comments at 10 (“Large load customers will use and benefit from the broader system and, therefore, should pay their fair share as users of that system rather than cover only certain costs.”); PJM TOs Comments at 3 (“Rolled-in rate treatment will promote more efficient and timely large load interconnection by avoiding cost allocation and jurisdictional disputes, as well as ensuring that all load customers continue to contribute their fair share of the benefits of an integrated transmission network.”); Comments of Entergy Operating Cos., Docket No. RM26-4 at 20 (filed Nov. 21, 2025) (“Entergy Comments”).

²⁷ EEI Comments, at 12-13.

²⁸ *Ameren Servs. Co. v. FERC*, 880 F.3d 571, 581 (D.C. Cir. 2018) (citing *Fed. Power Comm'n v. Hope Nat. Gas Co.*, 320 U.S. 591, 603 (1944)) (“But a careful reading of Supreme Court precedent reveals that a regulated industry is entitled to a return that is sufficient to ensure that new capital can be attracted.”); *Bluefield Waterworks & Imp. Co. v. Pub. Serv. Comm'n of W. Va.*, 262 U.S. 679, 690 (1923) (“There must be a fair return upon the reasonable value of the property at the time it is being used for the public.”) (citation omitted).

III. PROMISING SHORT-TERM SOLUTIONS FOR SPEED TO POWER

A. Several proposed speed-to-power solutions raised in the record are worth exploring

Several commenters proposed solutions addressing speed-to-power in the short term that would be worthwhile for the Commission to explore.²⁹ In particular, AEP highlights SPP’s HILL and HILLGA proposals and NRG’s “open season” concept as potential innovative and promising solutions that contain elements for the Commission to consider.

As several commenters note,³⁰ SPP’s HILL and HILLGA proposals serve as a useful example of RTO/ISO innovation to advance speed-to-power.³¹ HILL is a new category of load in SPP and HILLGA is a new generator interconnection service and process for generation supporting HILLs.³² These proposals shorten the study time for both large loads and additive, new generation resources to serve those loads. Reforms with elements modeled after these SPP proposals would work together with AEP’s vision of proactive transmission planning and entry-fee based generation interconnection. AEP’s proposals are aimed at efficient and cost-effective transmission planning, reducing reactive, piecemeal transmission projects, and reducing the timeline for generation to come online.³³ This goes hand-in-hand with consolidated planning ideas, discussed further herein, which may also use an entry fee model for generator interconnection requests.

²⁹ See, e.g., DTE Comments, at 2-3; Comments of Elec. Customer All., Docket No. RM26-4, at 2-3 (filed Nov. 21, 2025) (“ECA Comments”); SPP Comments at 4-6; NRG Comments at 17-26.

³⁰ See DTE Comments, at 8-10; Comments of Enchanted Rock, LLC, Docket No. RM26-4, at 4 (filed Nov. 21, 2025) (“Enchanted Rock Comments”); Google Comments at 8-9; Initial Comments of Mo. Pub. Serv. Comm’n, Docket No. RM26-4, at 4 (filed Nov. 21, 2025) (Missouri PSC).

³¹ See SPP Comments at 4-6, *see generally id.*

³² *Id.* at 4-5.

³³ See *generally* AEP Comments (describing “connect and manage” approach to generation interconnection, entry fee model, proactive transmission planning, and nationwide cost allocation).

NRG proposes an “open season” rather than a “first-come, first-served” process for large load interconnections.³⁴ Elements of that idea are also worth exploring.³⁵ An initial open season, in concept, could efficiently allocate existing headroom on the transmission system. An open season is also consistent with proactive planning, discussed in section IV, *infra*, because regular open seasons can support larger, more cost-effective transmission system expansion than studies that are focused on the individual needs of specific interconnection customers. Open seasons would lead to identification of electrically advantageous locations where load and generation can interconnect more efficiently and rapidly.³⁶ Additionally, an open season approach would help protect existing customers by requiring large load customers to enter into financially binding commitments to receive rights to interconnect with the system. Thus, this is a promising concept that the Commission should consider as it looks for ways to promote the development of the infrastructure necessary to support new large load interconnections.

Nonetheless, as discussed in section II, *supra*, when considering these or any other speed-to-power solutions, the Commission should avoid overly prescriptive requirements. Ultimately, regions need flexibility to adapt to their specific large load growth needs.

³⁴ NRG Comments at 17 – 26.

³⁵ See Initial Comments Elec. Power Supply Ass’n, Docket No. RM26-4, at 15 (filed Nov. 21, 2025) (“EPSA Comments”); Eolian Comments, at 32; Comments of ClearPath, Inc., Docket No. RM26-4, at 5 (filed Nov. 21, 2025) (“ClearPath Comments”).

³⁶ Here, “locations on the grid” as part of an open season concept differ from other mechanisms that seek to promote grid development, such as DOE National Interest Electric Transmission Corridors (“NIETC”). An open season concept is aimed at rapid load and generation development dictated by the engineering attributes of the grid, whereas NIETCs foster transmission expansion using static geographic zone designated by government studies.

B. As a bridge to firm service, the Commission should consider directing transmission providers to offer conditional service to generators and to LSEs serving large loads and co-located loads

The Commission should consider directing transmission providers to offer interim conditional service—that is, non-firm or conditional firm transmission service—to generators and to LSEs serving large loads and co-located loads. Transmission infrastructure investment is a barrier to firm service for large loads and generation. Temporary conditional service is a short-term speed-to-power solution that can bring new generation and load on line quickly and be used as a bridge service until the transmission system is upgraded to support full network service to those interconnections. This could advance large load interconnections without risking the reliability of the grid, which is paramount.

1. Conditional service for generation, large loads, and co-located loads could increase speed-to-power

Conditional generation interconnection service would address the root issue that large load growth faces: insufficient availability of reliable generation. Compared to firm service, conditional interconnection service could require less time for initial generation interconnection studies. Thus, it could improve speed-to-power for new generation needed to serve large loads and do so without threatening reliability for other customers. Conditional service also could facilitate a “connect and manage,” entry-fee approach to generation interconnection advocated for by AEP in its Initial Comments.³⁷ Using conditional interconnection service, generators could interconnect before full network service is available and the transmission provider could manage that service through curtailments until the necessary upgrades are in place to provide the generator with full network service.

³⁷ AEP Initial Comments, at 7-10.

Conditional transmission service could also provide a speed-to-power solution for large loads and co-located loads. If large loads are willing to accept conditional transmission service and be curtailed, the Commission could adopt rules that allow LSEs to bring large loads online faster. This approach would work under various scenarios such as standalone large loads that are willing to be curtailable or loads willing to co-locate with behind the meter generation, contract with nearby generation, or a combination thereof. In exchange for being curtailable, LSEs could give large loads the opportunity to come online faster.

AEP is already innovating to provide custom, behind-the-meter solutions to large load customers via its partnership with Bloom Energy. Under its agreement with Bloom Energy, AEP can secure solid oxide fuel cells to provide behind-the-meter power to large load customers quickly while the grid is being built out to accommodate their demand.³⁸ Whereas conditional service may require customers to be curtailable, AEP's partnership with Bloom Energy allows customers to come online sooner without curtailment. The development of conditional transmission service offerings would allow for more large load customers to consider novel, temporary flexibility solutions that improve speed-to-power without precluding an eventual transition to full network service.

Other commenters noted their support for similar approaches. Eolian detailed the benefits of flexible approaches for curtailable and hybrid facilities alike, describing how such approaches can reduce the resource adequacy impacts of large loads while enabling them to come online quickly.³⁹ Eolian also notes that without access to conditional transmission service, much of the

³⁸ AEP, *AEP Leveraging Fuel Cell Technology to Power Data Center Growth* (Nov. 14, 2024), <https://www.aep.com/news/stories/view/9866/AEP-Leveraging-Fuel-Cell-Technology-to-Power-Data-Center-Growth/>.

³⁹ *See generally*, Eolian Comments; *see id.* at 1

potential value of the hybrid resource model would be lost.⁴⁰ EsVolta highlighted that battery storage in particular can be deployed on a short, 12-18 month timeline.⁴¹ Advanced Energy United encourages the Commission to recognize that “[e]ven a small degree of flexibility among new large loads would enable the current system to accommodate significantly more load growth with minimal infrastructure upgrades.”⁴²

2. Any conditional service must be designed to ensure every customer does not threaten reliability and pays their fair share for use of the transmission system

While conditional generator interconnection and transmission service offerings could provide important benefits, as described above, it is essential that such services are designed to avoid threatening system reliability or allowing customers to escape paying their fair share for the services they receive. As the PJM Transmission Owners argued in their answer⁴³ to the Commission’s Co-Location Order to Show Cause,⁴⁴ even if co-located load rarely draws from the grid, transmission providers have an obligation to ensure the transmission system can serve the co-located load’s full needs at any time. Load benefits from its interconnection to the transmission system and the reliability that such interconnection provides, even co-located load that does not regularly draw from the grid.

⁴⁰ *Id.* 34.

⁴¹ Comments of EsVolta LP, Docket No. RM26-4, at 9 (filed Nov. 21, 2025) (“EsVolta Comments”).

⁴² Comments of Advanced Energy United, Docket No. RM26-4, at 22 (filed Nov. 21, 2025) (“Advanced Energy United Comments”) (citation omitted); *see also* Joint Comments of the Load Flexibility Parties, Docket No. RM26-4, at 2-4 (filed Nov. 21, 2025) (“Load Flexibility Parties”) (arguing that the Commission should consider flexible load eligible along with co-located load for expedited interconnection process).

⁴³ *PJM Interconnection L.L.C., et al.*, Answer of the Indicated PJM Transmission Owners to the Order Instituting the Proceeding, Accession No. 20250324-5298, Docket No. EL25-49, Attachment A at 13 (filed Mar. 24, 2025) (“PJM TO Co-Location OSC Answer”).

⁴⁴ *PJM Interconnection, L.L.C.*, 190 FERC ¶ 61,115 (2025).

As the demand and urgency for large load interconnection increases, AEP remains open to transmission providers offering conditional service to LSEs serving those loads, as long as the terms of such service are appropriately defined and reliability and existing customers is protected. In the long-term, co-located load must be served by existing forms of transmission service (Network Service or Point to Point Service), but specific forms of conditional service may be worth exploring as interim tools to help meet the unprecedented growth in large loads.⁴⁵ In particular, conditional transmission service could be an effective bridge solution to bring new large load customers online quickly, before the necessary transmission upgrades are made to provide the customer firm service, then the customer could convert to firm service once the upgrades are complete.⁴⁶ As noted above, the specific terms of any such conditional generation interconnection or transmission service are key—it cannot be a service that would permit any facility to interconnect to the grid without paying its fair share for the transmission system. But the potential speed-to-power benefits of conditional service could make it worthwhile to figure out these specifics and make conditional service workable for large load customers and associated generation without negatively impacting existing customers.

C. Short-term solutions are critical, but they cannot replace essential upgrades to the transmission system

Some commenters argued that short-term solutions such as co-location and curtailable large loads are on their own sufficient to allow for large load interconnection, mitigating the need

⁴⁵ PJM TO Co-Location OSC Answer, at 12.

⁴⁶ Large load customers, in particular data centers, typically seek Network Service. Conditional service is, therefore, only appropriate as a short-term tool to get these customers online more quickly so they do not have to wait for infrastructure. Then they can transition to Network Service.

for network upgrades.⁴⁷ This is not the case. Short-term, flexible approaches can facilitate speed-to-power, but they cannot ensure a transmission system that can support the projected large load growth. To meet the demands of large load growth, we must also expand generation capacity and transmission system capacity with proactive transmission planning.

IV. PROACTIVE TRANSMISSION PLANNING IS ESSENTIAL

A. Proactive transmission planning is also needed to ensure affordability and cost-effective system expansion

While short-term speed-to-power solutions are important, the Commission cannot overlook the need to foster proactive transmission planning, which is necessary to ensure cost-effective and affordable transmission system expansion and timely interconnection of new generation resources. In addition to AEP, many other commenters support proactive, multi-value planning as an essential tool to address large load interconnections. For example, Americans for a Clean Energy Grid commented that the ANOPR “misses one major point: the best way to integrate new facilities on the grid is to build out sufficient capacity through proactive well-planned multi-value transmission.”⁴⁸

⁴⁷ See Comments Emerald AI, Docket No. RM26-4, at 1 (filed Nov. 21, 2025) (“Emerald AI Comments”); Comments of the Inst. for Progress at 9-10 (filed Nov. 21, 2025) (“IFP Comments”).

⁴⁸ Comments of Am. for a Clean Energy Grid, Docket No. RM26-4, at 3-4 (filed Nov. 21, 2025) (“ACEG Comments”); *see also* Initial Comments of R Street Inst., Docket No. RM26-4, at 4 (filed Nov. 21, 2025) (“R Street Institute Comments”) (noting the Commission should “focus on forward-looking transmission development, rather than post-hoc network development accompanying generation interconnection requests.”); Comments of Pub. Int. Org., Docket No. RM26-4, at 20-21 (filed Nov. 21, 2025) (“Public Interest Organizations Comments”) (“[I]n the long term, building out the system in response to discrete, emergent needs rather than through proactive, forward-looking, system-wide planning fails to identify the most cost-effective infrastructure investments, which would solve the reliability issues associated with new large load *and* provide broader system benefits.”); Comments of the Am. Council on Renewable Energy, Docket No. RM26-4, at 3 (filed Nov. 21, 2025) (“American Council on Renewable Energy Comments”) (suggesting that the Commission work with DOE and the National Laboratories to “establish best practices and common methodologies for the forecasting of large loads and their power needs that can be incorporated into the long-term transmission planning process.”); Google Comments, at 3-4 (criticizing the current “fragmented, highly localized planning practices” and “piecemeal upgrades” and arguing for a “holistic”

As AEP noted in its comments on the Order No. 1920 rulemaking proceeding, forward looking, multi-value transmission planning goes beyond the “just in time” approach to transmission planning that is focused on addressing specific system needs as they occur.⁴⁹ Multi-value planning identifies regional solutions that address a multitude of needs that will occur over a long-term horizon, such as regional reliability and economic value. This creates more efficient and cost-effective solutions than the traditional “just in time” approach.⁵⁰ Proactive, multi-value planning is also crucial to foster the development of Extra High Voltage (“EHV”) facilities—the essential backbone of the nation’s transmission system—because it looks ahead to identify regional and interregional needs and the corresponding EHV solutions.

While Order No. 1920 took significant strides toward proactive transmission planning on a national scale, the Commission can take further steps to ensure that reforms adopted through compliance filings support large load growth. First, the Commission should facilitate expedient implementation of Order No. 1920’s planning mechanisms so that long-term, proactive transmission planning processes begin as soon as possible. Second, when considering compliance with Order No. 1920, the Commission should ensure that transmission providers’ scenario development processes include robust forecasting procedures for large loads that make sense for their region’s unique needs. While Order No. 1920 did not require specific forecasting procedures for large loads, it acknowledged that its reforms were driven, in part, by the changing demand

and “comprehensive, coordinated transmission planning process built on a more complete picture of growth in supply and demand.”).

⁴⁹ *Bldg. for the Future Through Elec. Reg’l Transmission Planning & Cost Allocation*, Initial Comments of Am. Elec. Power Serv. Corp., Docket No. RM21-17, at 12-15 (filed Oct. 12, 2021) (“AEP 1920 Comments”).

⁵⁰ *Id.* AEP supports the Commission’s efforts through the Order 1920 rulemaking proceeding to require long-term, multi-value regional transmission planning even though AEP has raised concerns about certain aspects of the Commission’s implementation of the rule which impede the Commission’s goals of encouraging efficient, cost effective, long-term development of grid infrastructure.

associated with large loads.⁵¹ Order No. 1920 also requires that long-term scenarios identify transmission needs across a plausible and diverse range of future scenarios, and the Commission should consider whether transmission providers' proposed approaches to scenario development adequately account for anticipated large load growth possibilities. Thus, the Commission would be acting within the requirements of Order No. 1920 if it ensured that transmission providers in their respective compliance filings consider large loads in their scenario development.

When done right, proactive planning could improve affordability when large loads join the system.⁵² Proactive planning identifies the most cost-effective transmission facilities and new large loads would share their portion of the fixed costs—a large portion—for such transmission investment. Together, these cost-effective investments paired with load that pays a proportionally large share of the fixed costs could reduce costs for other customers. For example, Pacific Gas and Electric Company noted that existing ratepayers could see bill reductions as a result of large loads interconnecting and contributing revenue to support the transmission system, because the revenues that exceed incremental interconnection costs can offset costs paid by existing ratepayers.⁵³ Also, as AEP noted in its Initial Comments, proactive planning will greatly increase

⁵¹ *Bldg. for the Future Through Elec. Reg'l Transmission Planning & Cost Allocation*, Order No. 1920, 187 FERC ¶ 61,068, at P 95 (2024); *subsequent history omitted*.

⁵² Ryan Wiser et al., *Factors influencing recent trends in retail electricity prices in the United States*, 38 *The Electricity Journal*, at 5-6 (2025) (providing analysis showing that load growth had the effect of depressing retail electricity rates by spreading fixed costs over greater load), <https://www.sciencedirect.com/science/article/pii/S1040619025000612>.

⁵³ Pac. Gas & Elec. Co. Opening Comments, Docket No. RM26-4, at 4-5 (filed Nov. 21, 2025) ("PG&E Comments"); *see also* Comments of Alliant Energy Corp. Servs., Docket No. RM26-4, at 3 (filed Nov. 21, 2025) ("Alliant Comments") ("Load growth, whether that is in the form of agricultural, technological, or manufacturing customers, creates an opportunity to maintain affordable rates for all customers as that growth helps to spread out fixed costs, support load predictability, and increase efficiency of grid assets."); ECA Comments, at 2 ("Effectively integrating large loads can also provide substantial benefits to all customers by spreading the fixed costs of the power system to the new large customers, leading to lower rates for existing customers.").

net benefits to consumers; a Grid Strategies paper found that delaying well-planned transmission increases costs and reduces benefits to consumers.⁵⁴

B. The Commission should support regional efforts for consolidated, integrated planning

The Commission should support regional efforts to move towards consolidated, integrated planning, like SPP's CPP.⁵⁵ As AEP noted in its Initial Comments, SPP's CPP embraces a version of an entry fee model for interconnection requests.⁵⁶ An entry fee model is designed to mitigate incentives for developers to submit speculative interconnection requests by providing greater cost certainty prior to entering the interconnection queue.⁵⁷ The SPP Transmission Owners Group believes that CPP will accomplish these goals, unanimously supporting the proposal.⁵⁸

As other comments have also urged, the Commission should look to SPP's CPP as a model of an integrated planning process for other regions to follow.⁵⁹ Consolidated planning allows interconnecting generation to contribute to regional transmission facilities being developed as part of a larger plan. Further, generators know their costs up front, and the interconnection process is

⁵⁴ AEP Initial Comments, at 17-18 & n.24 (citing Zach Zimmerman, Rob Gramlich & Michael Goggin, *Delaying transmission increases costs and reduced benefits for consumers*, at II (2025), <https://wiresgroup.com/wp-content/uploads/2025/11/Cost-of-Delayed-TransmissionReport-by-Grid-Strategies-for-WIRES.pdf>).

⁵⁵ See, e.g., Comments of Clean Energy Buyers Ass'n, Docket No. RM26-4, at 8-9 (filed Nov. 21, 2025) ("CEBA Comments"); Comments of the Am. Council on Renewable Energy, Docket No. RM26-4, at 4-5 (filed Nov. 21, 2025) ("ACORE Comments"); Comments of ENGIE N. Am., Inc., Docket No. RM26-4, at 5-6 (filed Nov. 24, 2025) ("ENGIE Comments"); Google Comments at 6-7; Comments of Southern Companies, Docket No. RM26-4, at 6-7 (filed Nov. 21, 2025) ("Southern Comments").

⁵⁶ AEP Initial Comments, at 12 n.15.

⁵⁷ See Sw. Power Pool, Inc., Submission of Revisions to the SPP Open Access Transmission Tariff to Implement the Consolidated Planning Process, Docket No. ER26-414-000 (filed Nov. 3, 2025) ("SPP CPP Filing").

⁵⁸ Comments of the Transmission Owners Grp., Docket No. ER26-414 (filed on Nov. 24, 2025) (SPP TOs Comments).

⁵⁹ See, e.g., Google Comments, at 4-5; Eolian Comments at 32; ClearPath Comments at 5.

streamlined by the decrease in speculative interconnection requests, which are often withdrawn. With fewer request withdrawals, the queue would not be so bogged down with restudies. Thus, consolidated planning provides a solution to speed up interconnection.

C. The Commission should encourage advanced conductors to maximize available infrastructure

The Commission should consider additional ways to encourage the use of advanced conductors to maximize the use of existing physical infrastructure and rights-of-way necessary to support the interconnection of large loads. Integrating advanced conductors in targeted applications can support the interconnection of large loads while new infrastructure is completed.⁶⁰ Reconductoring with advanced conductors can greatly increase capacity in a short time frame (less than two years), utilizing existing towers and rights-of-way, thereby avoiding siting processes.⁶¹ Thus, it provides a another tool the Commission should consider encouraging the use of to facilitate its goal of speed-to-power. Recently, AEP received a \$1.6 billion loan guarantee from DOE's Loan Programs Office (the Energy Dominance Financing Program) to rebuild and reconductor approximately 5,000 miles of transmission lines across Indiana, Michigan, Ohio, Oklahoma, and West Virginia.⁶² This financing will support upgrades to install advanced conductors, increasing

⁶⁰ For example, AEP Texas and ETT (Electric Transmission Texas) developed improvements to the transmission lines in the Lower Rio Grande Valley (LRGV) in Texas to improve reliability. AEP Texas used an advanced conductor for the LRGV Reconductor project. ETT had a concurrent greenfield project that was part of the same ERCOT endorsement, and the advanced conductor was used to expedite completion and maintain service to growing load in the LRGV while the greenfield project was being constructed.

⁶¹ See, e.g., Comments of CTC Glob., Docket No. RM26-4 (filed Nov. 24, 2025) ("CTC Global Comments"); ACORE Comments at 6-7; Initial Comments of the Working for Advanced Transmission Tech., Docket No. RM26-4 (filed Nov. 21, 2025); Public Interest Organizations Comments at 6.

⁶² DOE, *Energy Dep't Closes Loan Guarantee to Strengthen U.S. Grid Reliability* (Oct. 16, 2025), <https://www.energy.gov/articles/energy-department-closes-loan-guarantee-strengthen-us-grid-reliability#:~:text=Breadcrumb,capacity%20in%20rapidly%20growing%20areas>; AEP, *AEP Receives U.S. Dep't of Energy Loan Guarantee to Upgrade 5,000 Miles of Transmission Lines* (Oct. 16, 2025), <https://www.aep.com/news/stories/view/10501/>.

capacity within existing rights-of-way. The Commission should likewise consider incentives to promote the widespread use of advanced conductors. This technology is a forward-looking tool to increase transmission capacity that also supports speed-to-power in the short term.

V. CONCLUSION

AEP urges the Commission to focus its actions on the long-term capacity additions and transmission system expansions that are necessary to support large load growth. The nation cannot keep pace with the technological and national security demands without improved generation interconnection processes, transmission planning, and further development of the backbone transmission system to support both. There are certain short-term, speed-to-power solutions that the Commission should explore in tandem with long-term, forward-looking generation interconnection, proactive transmission planning, and cost allocation solutions. Finally, the Commission's reforms should avoid introducing additional delays to large load interconnection by disrupting state and regional-level efforts.

Respectfully submitted,

/s/ John C. Crespo

John C. Crespo

Carrie L. Bumgarner

Jessica A. Cano

American Electric Power Service Corporation

1 Riverside Plaza

Columbus, OH 43215

Telephone: (614) 716-3727

E-mail: jccrespo@aep.com

Telephone: (614) 716-2941

E-mail: clbumgarner@aep.com

Telephone: (614) 716-2921

E-mail: jacano@aep.com

December 5, 2025

CERTIFICATE OF SERVICE

I hereby certify that on this 5th day of December 2025, I have served a copy of the foregoing document on the official service list compiled by the Office of the Secretary for the above-referenced proceeding.

/s/ LaChon Turner

LaChon Turner

Paralegal

American Electric Power Service Corporation

801 Pennsylvania Ave NW

Suite 735

Washington, DC 20004

(202) 383-3465

lturner@aep.com